Superfluid Cooling Feedback as the Origin of Dark Energy and Dark Matter

Erik Otto

Working Paper in the Dynamic Aether Mechanics Framework

Abstract

Within the Dynamic Aether Mechanics framework, a single mechanism—the freeze-out of roton-like excitations as the cosmic Aether cools—is shown to produce both dark energy and dark matter through a self-reinforcing feedback loop. As the universe expands, the Aether temperature drops below the roton energy gap, exponentially suppressing the normal-fluid fraction. This collapse in normal fraction eliminates the bulk viscosity that had been braking cosmic expansion, creating an irreversible ratchet: expansion causes cooling, cooling reduces viscous resistance, reduced resistance permits faster expansion. The mechanism naturally explains the onset of cosmic acceleration ~5 Gyr ago (the epoch when the Aether temperature crossed below the roton gap), the phantom-to-quintessence crossing observed by DESI, and the near equality of the CMB temperature and the transition temperature. Early-universe dark matter effects arise through a complementary channel: the temperature-dependent gravitational asymmetry fraction $\epsilon(T)$ produces stronger effective gravity in cold, dense regions, mimicking the gravitational signatures conventionally attributed to dark matter particles. The two dark phenomena are thus revealed as dual consequences of the same underlying thermodynamic transition—resolving the cosmic coincidence problem.

This working paper explores a specific mechanism within the Dynamic Aether Mechanics framework. It references results from Paper 1 (General Overview), Paper 3 (Gravity), and Paper 6 (Cosmology) of the main series. The ideas presented here are under development and may be incorporated into the main papers in revised form.

1. Introduction

Modern cosmology faces two deeply puzzling observations: approximately 68% of the universe's energy budget consists of dark energy—a repulsive influence causing the expansion of the universe to accelerate—while approximately 27% consists of dark matter—an attractive influence that binds galaxies and clusters more tightly than their visible mass can account for. The standard ΛCDM model treats these as entirely separate phenomena: a cosmological constant Λ for dark energy and cold, collisionless particles for dark matter.

This separation raises an uncomfortable question known as the *cosmic coincidence problem*: why are the densities of dark energy and dark matter similar in magnitude *right now*? In ΛCDM, dark energy density is constant while matter density dilutes as a^{-3}; their near-equality today is a fleeting accident with no physical explanation.

The Dynamic Aether Mechanics framework (Papers 1–7) offers a different perspective. In this framework, both dark energy and dark matter arise from thermodynamic properties of a single medium—the Aether—specifically from its two-fluid (superfluid + normal) structure. Paper 6 identifies dark energy as the thermal pressure of hot void Aether, sustained by bulk-viscous dissipation of the Hubble flow, and dark matter as the two-fluid first-sound correction that produces additional centripetal acceleration through the acoustic metric.

This working paper develops a specific mechanism—*roton freeze-out*—that connects these two phenomena through a single thermodynamic transition and explains their simultaneous emergence. The central idea is that the Aether's excitation spectrum includes a roton-like energy gap Δ, and as the cosmic temperature drops below Δ/k_B, the resulting exponential suppression of the normal fraction triggers a cascade of consequences that manifest as both dark energy and dark matter.

2. The Cooling Feedback Mechanism

2.1 Global Cooling of the Free Aether

Two processes have shaped the thermal state of the Aether over cosmic history (Paper 6, Section 1):

1. **Permanent binding** locks thermal energy into mass. Each mass-formation event (driven by the Dynamic Casimir Effect) extracts energy from the local Aether, cooling the matter-containing regions.
2. **Stellar radiation** heats the Aether in voids, partially offsetting the cooling. However, the integrated stellar radiation ($\sim 10^{-14}$ J/m^3) falls $\sim 10^4$ times short of the observed dark energy density, so the dominant energy channel is bulk-viscous dissipation of the Hubble flow.

The net effect over cosmic time is a gradual decrease in the mean Aether temperature. The CMB temperature $T_{\mathrm{CMB}}(z) = T_0(1 + z)$, where $T_0 = 2.725$ K, tracks this cooling: at redshift $z = 2$ the Aether was at ~ 8 K, and at $z = 5$ it was at ~ 16 K.

2.2 Roton-Like Excitations and the Energy Gap

In superfluid helium-4, the excitation spectrum contains two branches: low-energy phonons ($\omega = ck$) and higher-energy rotons with a characteristic energy gap $\Delta_{\text{He}}/k_B \approx 8.6$ K. Below this gap, roton excitations are exponentially suppressed, and the normal fraction collapses:

$$\frac{\rho_n}{\rho_A} \propto \exp\left(-\frac{\Delta}{k_B T}\right) \quad \text{for} \quad T \lesssim \Delta/k_B \tag{1}$$

If the Aether possesses analogous intermediate-wavelength excitations (an open question identified in Paper 6, Section 1.10), the same exponential suppression occurs as the cosmic temperature drops below the gap energy. The roton gap Δ/k_B for the Aether need not equal helium's value; it is determined by the Aether's specific interaction parameters (m_A, ρ_A, K_A).

Note that the full Landau formula for helium-4 rotons includes a prefactor $\rho_{n,\text{roton}} \propto T^{-1/2} \exp(-\Delta/k_B T)$. The $T^{-1/2}$ factor is specific to helium's roton dispersion relation; the Aether's excitation spectrum may produce a different prefactor. The exponential Boltzmann suppression is universal for any gapped excitation and dominates the temperature dependence. Throughout this paper, the pure exponential is used as an approximation; the prefactor introduces corrections of order $(1 + z)^{-1/2}$ (a ~40% reduction at $z = 2$) that do not qualitatively change the results.

Paper 6 already identifies the excitation spectrum as the critical unknown that simultaneously determines the dark matter magnitude ($\Delta c_1^2/c^2 \sim 10^{-6}$), the bulk viscosity ($\zeta_n \sim 10^6$ Pa s), the void temperature ratio ($T_{\text{void}}/T_{\text{wall}} \approx 2.5$), and the dark energy equation of state $w(z)$. This paper proposes that the roton gap Δ is the single parameter that governs all four.

2.3 The Self-Reinforcing Ratchet

The cooling feedback operates through a self-reinforcing cycle:

1. **Expansion does adiabatic work.** The Hubble flow cools the normal component of the Aether (the superfluid component, carrying zero entropy, cannot be cooled further).

2. **Cooling suppresses roton excitations.** As T drops further below Δ/k_B, roton modes freeze out exponentially, converting normal fluid to superfluid.

3. **The feedback becomes self-sustaining.** In principle, heating can reconvert superfluid to normal fluid (as in helium-4). However, the viscous heating that could replenish the normal fraction is itself *proportional to the normal fraction* ($\zeta_n \propto \rho_n$). Once ρ_n drops below a critical threshold, the heating rate becomes too weak to offset the cooling from expansion—the feedback loop becomes supercritical. From this point, cooling outpaces heating, the normal fraction continues

to fall, and the process is dynamically irreversible: a self-reinforcing instability rather than a thermodynamic one-way door.

4. **Viscosity collapses.** Bulk viscosity ζ_n is carried exclusively by the normal component. As the normal fraction drops, so does ζ_n, reducing the viscous braking that had been decelerating the expansion.

5. **Expansion accelerates.** With less viscous resistance, the existing expansion—driven by the thermal pressure differential between voids and matter regions—encounters less opposition. The expansion rate increases.

6. **Return to step 1.** Faster expansion produces more adiabatic cooling, driving further roton freeze-out.

This is a positive feedback loop that becomes self-sustaining once the normal fraction drops below the critical threshold (step 3). Below this threshold, the ratchet only tightens: the system cannot return to the high-viscosity, braking-dominated regime because the mechanism that would restore it (viscous heating) has been weakened by the very transition it would need to reverse. The result is an accelerating expansion that we observe as dark energy.

3. Dark Energy from Roton Freeze-Out

3.1 Viscous Heating Rate

Paper 6 derives the bulk-viscous dissipation rate from the Hubble flow:

$$\dot{\varepsilon} \;=\; 9\,\zeta_n\,H^2 \tag{2}$$

In the roton freeze-out model, the bulk viscosity evolves with redshift through the normal fraction:

$$\zeta_n(z) \;=\; \zeta_0\,f_n(z) \tag{3}$$

where $\zeta_0 \sim 10^6$ Pa s is the present-day value and the relative normal fraction is:

$$f_n(z) \;\equiv\; \frac{\rho_n(z)}{\rho_n(0)} \;=\; \exp\!\left[\frac{\Delta}{k_B T_0}\,\frac{z}{1+z}\right] \tag{4}$$

This follows from Eq. 1 with $T(z) = T_0(1+z)$:

$$f_n(z) = \frac{\exp(-\Delta/[k_B T_0(1+z)])}{\exp(-\Delta/[k_B T_0])} = \exp\!\left[\frac{\Delta}{k_B T_0}\left(1 - \frac{1}{1+z}\right)\right] = \exp\!\left[\frac{\alpha\,z}{1+z}\right] \tag{5}$$

where $\alpha \equiv \Delta/(k_B T_0)$.

3.2 Normal Fraction and Heating Rate Tables

For $\Delta/k_B = 6$ K (constrained below by the DESI crossing redshift), $\alpha = 6/2.725 = 2.20$. The resulting evolution:

Redshift z	T_{CMB} (K)	$f_n(z)$	$(H/H_0)^2$	$S(z)/S(0)$
0	2.73	1.00	1.00	1.0
0.3	3.54	1.66	1.36	2.3
0.5	4.09	2.08	1.71	3.6
0.7	4.63	2.48	2.17	5.4
1.0	5.45	3.01	3.10	9.3
2.0	8.18	4.34	8.80	38.2
5.0	16.4	6.26	65.5	410

Table 1: Evolution of the normal fraction ratio $f_n(z)$, the squared Hubble parameter, and the total viscous heating rate $S(z) = 9\zeta_n(z)H(z)^2$ relative to today, for a roton gap $\Delta/k_B = 6$ K. The Hubble parameter uses $H(z) = H_0\sqrt{0.3(1+z)^3 + 0.7}$.

The viscous heating rate at $z = 1$ was approximately 9× stronger than today, and at $z = 2$ it was approximately 38× stronger. This dramatic change is driven primarily by the exponential roton factor at moderate redshifts and by the H^2 factor at higher redshifts.

3.3 Phantom–Quintessence Crossing

The effective dark energy equation of state parameter $w(z)$ is defined by how the dark energy density ρ_{DE} evolves:

$$w(z) \;=\; -1 \;-\; \frac{1}{3H}\,\frac{d\ln\rho_{\mathrm{DE}}}{dt} \tag{6}$$

The dark energy density evolves through two competing effects:

- **Viscous heating** (adds energy): $\dot{\varepsilon}_{\mathrm{heat}} = 9\,\zeta_n(z)\,H^2$
- **Expansion dilution** (removes energy): $\dot{\varepsilon}_{\mathrm{dilute}} \propto H\,P_{\mathrm{th}}$ (see Section 4 for discussion of the dilution rate)

The balance determines the sign of $d\rho_{\mathrm{DE}}/dt$ and hence whether w is above or below -1:

- When heating dominates ($S > D$): ρ_{DE} is increasing $\Rightarrow w < -1$ **(phantom)**

- When dilution dominates ($D > S$): ρ_{DE} is decreasing $\Rightarrow w > -1$ **(quintessence)**
- At the crossover ($S = D$): $w = -1$ exactly

At high redshift, the viscous heating rate was far stronger than today (Table 1), so heating dominated—placing the early dark energy firmly in the phantom regime ($w < -1$). As the Aether cooled through the roton gap, the normal fraction collapsed exponentially, viscous heating dropped, and dilution began to dominate—transitioning to the quintessence regime ($w > -1$).

This is precisely the phantom-to-quintessence crossing observed by DESI at 2.8–4.2σ significance [1].

4. The $w(z)$ Equation of State

4.1 Evolution of the Dark Energy Density

The thermal pressure P_{th} of the void Aether evolves through two competing effects:

$$\frac{dP_{\mathrm{th}}}{dt} = \underbrace{9\,\zeta_n(z)\,H^2}_{\text{viscous heating}} - \underbrace{\beta\,H\,P_{\mathrm{th}}}_{\text{expansion work}} \tag{7}$$

where β parameterizes the rate at which the thermal pressure does work on the expansion. In a standard expanding fluid, $\beta = 3$ (matter-like dilution $\propto a^{-3}$). However, the Aether density is held essentially constant ($\delta\rho_A/\rho_A \sim 10^{-4}$) by the enormous bulk modulus $K_A \sim 10^{28}$ J/m^3, so the thermal energy density does not dilute in the usual sense. The effective β depends on how efficiently the thermal pressure converts to kinetic energy of expansion and is an open calculation within the Aether framework.

Why the standard $w(z)$ mapping is non-trivial. In the Aether model, dark energy is not a homogeneous negative-pressure fluid as in ΛCDM. It is the *thermal pressure gradient* between hot voids and cold matter-containing regions that drives galaxy recession—a real force, not a geometric effect. The equation of state parameter $w(z)$ is an *effective* quantity extracted when observers fit the expansion history $H(z)$ to the ΛCDM framework. Deriving $w(z)$ precisely requires computing $H(z)$ from the Aether's pressure-gradient dynamics and then fitting it to the standard parameterization—a calculation that depends on β and the full thermal history.

Robust qualitative prediction. Despite the mapping complexity, the *direction* of the phantom–quintessence crossing is robust and does not depend on β. The viscous heating rate $S(z) = 9\,\zeta_n(z)\,H^2$ was dramatically stronger in the past (Table 1). Regardless of the dilution rate, this means:

- At high z: strong viscous heating was building up the effective dark energy density faster than dilution could reduce it $\Rightarrow$ when fit to ΛCDM, this appears as $w < -1$ **(phantom)**.

- At low z: roton freeze-out has collapsed the heating rate, so whatever dilution exists now dominates $\Rightarrow$ this appears as $w > -1$ **(quintessence)**.

- The crossover from one regime to the other is the phantom–quintessence crossing observed by DESI.

4.2 Constraining the Roton Gap from DESI

The phantom–quintessence crossing occurs at the redshift z_* where the viscous heating rate equals the expansion dilution rate (Eq. 7 with $dP_{\text{th}}/dt = 0$). The crossing redshift depends on $\alpha = \Delta/(k_B T_0)$ through $\zeta_n(z) = \zeta_0 \exp[\alpha z/(1+z)]$. Since ζ_n grows exponentially with z while the dilution rate grows more slowly, the crossing moves to higher z for smaller α (smaller roton gap—slower exponential growth, so the balance point shifts outward).

Δ/k_B (K)	α	Estimated z_*
4	1.47	~1.5
5	1.83	~1.0–1.2
6	2.20	~0.7–1.0
8	2.94	~0.3–0.5
12	4.40	~0.1–0.2

Table 2: Estimated phantom–quintessence crossing redshift z_* as a function of the roton gap Δ/k_B. DESI observations indicate $z_* \gtrsim 1$, constraining $\Delta/k_B \approx 5$–6 K.

DESI's observed crossing at $z \gtrsim 1$ constrains the roton gap to $\Delta/k_B \approx 5$–6 K. This is modestly below the helium-4 roton gap (8.6 K), consistent with the fact that the Aether has fundamentally different microscopic parameters ($m_A \sim 700$ MeV/c^2 vs. $m_{\text{He}} = 4$ amu; $\rho_A \sim 10^{12}$ kg/m^3 vs. $\rho_{\text{He}} \sim 145$ kg/m^3).

4.3 Qualitative Shape of $w(z)$

The model predicts a distinctive $w(z)$ shape:

- At $z \gg z_*$: viscous heating strongly dominates $\Rightarrow$ $w \ll -1$ (deep phantom)

- Near $z \approx z_*$: heating and dilution balance $\Rightarrow$ w crosses -1

- At $z = 0$: dilution dominates $\Rightarrow$ $w > -1$ (quintessence)

- Far future ($z < 0$): roton freeze-out essentially complete, $\zeta_n \to 0 \Rightarrow w$ approaches a value determined by the dilution rate β (no viscous heating to offset dilution)

Crucially, $w(z)$ follows a *phase-transition curve* (exponential in Δ/T), not the smooth polynomial assumed by the CPL parameterization $w(a) = w_0 + w_a(1 - a)$. Future high-precision $w(z)$ measurements could distinguish between the exponential form predicted here and polynomial models.

4.4 The CMB Temperature Coincidence

The current CMB temperature ($T_0 = 2.725$ K) lies within a factor of two of the constrained roton gap ($\Delta/k_B \approx 5$–6 K). This is not a coincidence—it is a *consequence* of the fact that we observe the universe during the transition epoch. Dark energy became observationally significant when the normal fraction dropped enough for the feedback ratchet to dominate. We detect it now precisely because T_0 is in the range where roton freeze-out is actively occurring.

The "coincidence problem" of ΛCDM—why is dark energy dominant *now*?—becomes a phase-transition timing question with a physical answer: the Aether crossed Δ/k_B in the recent cosmological past, and the resulting ratchet is still tightening.

5. Dark Matter from Varying G

5.1 Temperature Dependence of the Gravitational Constant

Paper 3 derives the gravitational constant as $G \propto \epsilon\, \rho_A\, \psi^2$, where ϵ is the *asymmetry fraction*—the fraction of each transient bind–unbind cycle's energy that escapes as the gravitational wave. Since the Aether density ρ_A is essentially constant ($\delta\rho_A/\rho_A \sim 10^{-4}$), the primary channel for G variation is through $\epsilon(T)$:

> Higher temperature provides thermal energy that resists binding, potentially reducing the fraction of each cycle's energy that escapes as the gravitational wave. Cold environments (near matter) could have higher ϵ and therefore stronger gravitational output, while hot environments (cosmic voids) could have lower ϵ. (Paper 3, Section 1.3)

This means:

$$G(T) \;=\; G_0\, \frac{\epsilon(T)}{\epsilon(T_0)} \tag{8}$$

where G_0 is the locally measured value (in cold, matter-containing regions).

5.2 Magnitude of the Variation

For structure formation, the dark matter mass-to-visible-mass ratio is approximately 5–6:1. If varying G is responsible, we need:

$$\frac{G_{\text{dense}}}{G_{\text{void}}} = \frac{\epsilon(T_{\text{matter}})}{\epsilon(T_{\text{void}})} \approx 6 \tag{9}$$

With the void-to-matter temperature ratio $T_{\text{void}}/T_{\text{matter}} \approx 2.5$ (Paper 6, Eq. 18), this requires:

$$\epsilon(T) \propto T^{-n}, \qquad 2.5^n = 6 \Rightarrow n \approx 1.95 \tag{10}$$

A near-quadratic temperature dependence $\epsilon \propto T^{-2}$ is physically motivated: each transient binding event involves a *pair* of particles (particle–antiparticle pair from the Dynamic Casimir Effect), and the thermal disruption of each member's binding efficiency contributes independently, giving a product of two temperature-dependent factors.

5.3 Consistency with Local G Measurements

The variation $G \propto T^{-2}$ does not conflict with the exquisite precision of local G measurements. All terrestrial and solar-system measurements of G occur in the same thermal environment—cold, matter-containing regions where $T \approx T_{\text{matter}}$ is essentially uniform. The measured value is $G(T_{\text{matter}})$, which is constant across all local experiments.

The *spatial* variation of G between galaxy cores and cosmic voids ($\Delta G/G \sim 5$–6) is not directly measured, because:

- Gravitational measurements (rotation curves, lensing) are made *near matter*, where $G = G_0$.
- What is conventionally attributed to "dark matter mass" is the *difference* between the gravitational effect produced by G_{dense} and what would be expected from G_{void} applied to the same visible mass.
- No experiment directly compares G at two locations with different Aether temperatures.

5.4 Constraints from the CMB

At the CMB epoch ($z \sim 1100$, $T \approx 3000$ K), the Aether temperature was nearly uniform: $\delta T/T \sim 10^{-5}$. The corresponding G variation:

$$\frac{\delta G}{G} = n\frac{\delta T}{T} \approx 2 \times 10^{-5} \tag{11}$$

This is a 0.002% variation—well within current CMB constraints on G variability, which are at the percent level. The effect would produce subtle modifications to the CMB high-ℓ power spectrum (enhanced Silk damping in slightly warmer regions), potentially detectable by next-generation polarization experiments (CMB-S4, LiteBIRD).

6. Mechanism Handoff: Early to Late Universe

The superfluid cooling model predicts that the *mechanism* underlying "dark matter" effects changes smoothly over cosmic time, even though the observational signatures remain broadly similar.

Epoch	z	T_{CMB} (K)	Primary DM mechanism	Primary DE mechanism
Recombination	∼1100	∼3000	First-sound correction	Viscous braking (strong)
Structure formation	2–20	8–55	First-sound + varying G	Braking (strong)
Transition	0.5–2	4–8	Both strengthening	Braking weakening
Present	0	2.7	First-sound (peak)	Ratchet engaged
Future	<0	<2.7	Weakening	Ratchet complete

Table 3: The dominant dark matter and dark energy mechanisms at different cosmic epochs. The first-sound correction (Paper 6) operates at all epochs. Varying G becomes a significant *supplementary* channel once structures form and temperature contrasts develop ($z \lesssim 20$).

Recombination ($z \sim 1100$, $T \sim 3000$ K). At the CMB epoch, the universe is nearly homogeneous ($\delta T/T \sim 10^{-5}$), so varying G is negligible ($\delta G/G \sim 2 \times 10^{-5}$). Dark matter effects at this epoch—reflected in the CMB acoustic peak heights—arise from the first-sound correction (Paper 6), which operates whenever a temperature gradient exists between denser and less dense regions. The derivation of the CMB power spectrum from first-sound dynamics remains an open calculation (Paper 6, Section 1.10).

Structure formation ($z \sim 2$–20, $T \sim 8$–55 K). As proto-structures collapse, they cool the local Aether through binding, creating growing temperature contrasts with the surrounding medium. This bootstrapping process progressively enhances G in dense regions: slight cooling → slightly stronger G → faster collapse → more cooling. By $z \sim 2$–3, significant structures exist with $T_{\mathrm{void}}/T_{\mathrm{matter}} \approx 2.5$, and the varying-$G$ channel contributes substantially alongside the first-sound correction.

Transition ($z \sim 0.5$–2, $T \sim 4$–8 K). The Aether is cooling through the roton gap. The normal fraction is dropping exponentially. Both varying G (still significant where temperature contrasts persist) and the first-sound correction (strengthening as the superfluid fraction increases near matter while voids retain more normal component) contribute to dark matter effects. Dark energy is "turning on" as viscous braking collapses.

Present ($z \approx 0, T \approx 2.7$ K). The normal fraction is low globally. Near matter (cold regions), the Aether is nearly pure superfluid. In voids (warmer), some normal component persists. The contrast in c_1 between these regions is at its peak, making the first-sound correction the dominant dark matter mechanism for galaxy rotation curves and gravitational lensing. Dark energy is fully engaged—the ratchet is tightening.

Far future ($z < 0, T < 2.7$ K). Eventually, even the voids go fully superfluid. The c_1 gradient flattens, weakening dark matter effects. G variation also weakens (temperatures converge). Galaxies gradually become less bound. Expansion continues accelerating (the ratchet is self-sustaining), heading toward a fully superfluid cosmos.

7. Observational Consistency Checks

7.1 DESI Dark Energy Equation of State

Observation: $w_0 > -1$ and $w_a < 0$ at 2.5–3.9σ; phantom-to-quintessence crossing at $z \gtrsim 1$ [1].

Model prediction: At $z = 0$, dilution dominates over viscous heating ($w > -1$, quintessence). Going back in time, the exponentially larger ζ_n makes heating dominate ($w < -1$, phantom). With $\Delta/k_B \approx 5$–6 K, the crossing occurs at $z \approx 0.7$–1.0.

Status: Consistent.

7.2 Bulk Viscosity

Observation: Bulk viscous cosmology models require $\zeta_0 \sim 10^6$ Pa s [2].

Model prediction: $\zeta_0 \sim 10^6$ Pa s (Paper 6, Eq. 2).

Status: Exact match.

7.3 JWST High-Redshift Galaxies

Observation: Unexpectedly massive, mature galaxies at $z > 6$. No clear evidence for modified dark matter properties, though results are consistent with both CDM and WDM.

Model prediction: At $z > 6$, the first-sound correction provides baseline dark matter effects, while varying G (enhanced in cold, dense regions via the bootstrapping process of Section 6) provides additional gravitational enhancement. Together, these produce *faster* structure formation than standard gravity alone. Massive galaxies forming early is a natural consequence. The gravitational signatures are indistinguishable from CDM because both "more mass" and "stronger G" produce the same observables (deeper potential wells, stronger lensing).

Status: Consistent; potentially explanatory.

7.4 CMB Constraints on Varying G

Observation: High-ℓ CMB power spectrum constrains G variations at the percent level.

Model prediction: At the CMB epoch ($z \sim 1100$, $T \approx 3000$ K), temperature variations are $\delta T/T \sim 10^{-5}$. The corresponding $\delta G/G \approx 2 \times 10^{-5}$—three orders of magnitude below current constraints.

Status: No conflict.

7.5 Dark Matter Halo Evolution

Observation: Halo concentrations at $z \sim 3$ are roughly constant across mass ranges ($c \sim 3.5$–4), with weak mass dependence that strengthens at lower z.

Model prediction: At $z \sim 3$, the dark matter effect includes both the first-sound correction and a growing contribution from G variation. The G-variation component is temperature-dependent (not mass-dependent), producing weak mass dependence. At $z \sim 0$, the first-sound correction dominates and depends on the galaxy's luminosity (which sets the temperature profile), introducing stronger mass dependence. The transition from weak to strong mass dependence is qualitatively consistent with the observed trend.

Status: Consistent.

7.6 Cosmic Coincidence

Problem: Why are $\Omega_{\rm DE}$ and $\Omega_{\rm DM}$ similar in magnitude today?

Resolution: Both phenomena are consequences of the same thermodynamic transition. Dark energy "turns on" when the normal fraction drops enough for the ratchet to engage. The first-sound dark matter correction reaches its maximum strength during the same transition (maximum contrast between matter and void superfluid fractions). Their co-emergence from a single event naturally produces similar magnitudes without fine-tuning.

Status: Resolved in principle.

8. Predictions

1. **Roton gap $\Delta/k_B \approx$ 5–6 K.** This is a specific, testable prediction. If the Aether's excitation spectrum can be derived from first principles (from m_A, ρ_A, K_A), the resulting roton gap must fall in this range. This same gap must simultaneously produce the correct dark matter magnitude, bulk viscosity, and void temperature ratio.

2. **$w(z)$ follows a phase-transition curve.** The equation of state is not a smooth polynomial but contains the exponential factor $\exp[\Delta/(k_B T_0) \cdot z/(1+z)]$. High-precision $w(z)$ measurements from future DESI data releases, Euclid, and the Vera Rubin Observatory could distinguish this shape from the CPL parameterization.

3. **Dark matter signatures evolve with redshift.** Specifically, the ratio of "dark" to visible mass should show subtle redshift dependence: at $z > 2$ (where G variation contributes significantly alongside the first-sound correction), dark matter effects may exhibit different scaling relations than at $z < 1$ (where the first-sound correction alone dominates). JWST rotation curve measurements at $z \sim 2$–3 could test this.

4. **Void regions are "further along" the transition.** Voids with lower temperatures should show stronger expansion rates and different dark matter characteristics than regions near large structures. This prediction is testable with upcoming void catalogs from DESI and Euclid.

5. **Late-time viscosity decrease.** The bulk viscosity of the cosmic medium should be measurably lower at $z = 0$ than at $z = 1$. This could manifest as redshift-dependent damping of baryon acoustic oscillations or gravitational wave propagation.

6. **Massive early galaxies from enhanced G.** The model predicts that structure formation was accelerated by G enhancement in cold, dense regions. This provides a natural explanation for the JWST observations of unexpectedly massive galaxies at $z > 6$, without requiring exotic dark matter properties or modified initial conditions.

9. Open Questions

1. **Functional form of $\epsilon(T)$.** The assumption $\epsilon \propto T^{-2}$ is motivated by the pair nature of transient binding, but a first-principles derivation from the Dynamic Casimir Effect threshold conditions is needed. Alternative forms (exponential suppression, logarithmic dependence) would change the magnitude and scaling of the varying-G dark matter effect.

2. **Derivation of the roton gap from Aether parameters.** Can $\Delta/k_B \approx$ 5–6 K be derived from the known Aether parameters ($m_A \approx$ 500–850 MeV/c^2, $\rho_A \approx 10^{11}$–10^{12} kg/m^3, $K_A \approx 10^{28}$–10^{29} J/m^3)? In helium-4, the roton gap arises from short-range correlations in the liquid

structure; the analogous calculation for the Aether requires knowledge of the inter-quasiparticle potential.

3. **Numerical $w(z)$ solution.** Computing the effective $w(z)$ requires two steps: (a) determine the dilution parameter β from the Aether's constant-density thermodynamics, and (b) solve Eq. 7 numerically with the evolving $\zeta_n(z)$ and $H(z)$ to produce $H(z)$, then fit the result to the CPL parameterization to extract w_0 and w_a. Overlaying this with DESI data would provide a quantitative test and refine the constraint on Δ/k_B.

4. **CMB power spectrum.** How does the varying-G mechanism at $z \sim 1100$ affect the acoustic peaks? The $\delta G/G \sim 10^{-5}$ variation is small but could produce detectable modifications to the damping tail.

5. **Rotation curves at different redshifts.** The handoff from G-variation to first-sound correction should produce measurable differences in rotation curve shapes at $z \sim 2$–3 versus $z \sim 0$. Quantitative predictions require the three-dimensional temperature profile $T(r, z)$ at different cosmic epochs.

6. **Reconciliation with existing Paper 6 framework.** The thermal pressure mechanism (Paper 6) and the roton freeze-out mechanism (this paper) are complementary: thermal pressure provides the driving force for expansion, while roton freeze-out explains why the expansion *accelerates*. A unified treatment should demonstrate that both effects arise naturally from the same two-fluid dynamics.

References

[1] DESI Collaboration, "DESI DR2 Results: Measurement of the Baryon Acoustic Oscillation scale and Hubble parameter across 11 billion years," *arXiv:2503.14738* (2025).

[2] A. Hernández-Almada *et al.*, "Bulk viscous cosmological models with a cosmological constant: Observational constraints," *arXiv:2508.11614* (2025).

[3] E. Otto, "Unifying Theory of Dynamic Aether Mechanics: General Overview," this series, Paper 1.

[4] E. Otto, "Unifying Theory of Dynamic Aether Mechanics: Gravity," this series, Paper 3.

[5] E. Otto, "Unifying Theory of Dynamic Aether Mechanics: Cosmology," this series, Paper 6.